

# LYA CHIÑAS (SHE/HER)

PHD RESEARCHER | MACHINE LEARNING FOR BIOLOGY

📍 Toronto, Canada

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🌐 [Lya Chiñas Serrano](#)

## EDUCATION

### Ph.D. STUDENT

University of Toronto,  
Dept. of Chemical  
Engineering

Toronto, Canada • 2024  
– Pres.

### B.S. IN BIOTECHNOLOGY ENGINEERING

Tecnológico de  
Monterrey

Puebla, Mexico • 2020 –  
2024

GPA 4.0, First in Class  
Honors

## SKILLS

**Languages:** Python,  
MATLAB, HTML, CSS

**Tools:** PyTorch,  
TensorFlow, scikit-learn,  
NumPy, Pandas,  
Matplotlib, Git,  
Bash/Shell, Slurm,  
LaTeX

**Content Creation:**  
Adobe (Photoshop,  
Illustrator, Premiere Pro),  
Affinity (Photo, Designer,  
Publisher)

## LANGUAGES

**Spanish** • Native

**English** • Advanced •  
TOEFL C1

**French** • Elementary •  
DELTA A2

## FELLOWSHIPS

Massey College Junior  
Fellow

Toronto, Canada • Sept  
2024 – Present

## EXPERIENCE

### GRADUATE STUDENT | LABORATORY FOR METABOLIC SYSTEMS ENGINEERING

University of Toronto • Sept 2024 – Pres.

- Creating and evaluating deep learning models for the prediction of enzyme kinetic parameters and *E. coli* phenotypic behavior.
- Building end-to-end ML pipelines, with multi-omics data compilation, model training, and evaluation.

### TEACHING ASSISTANT | UNIVERSITY OF TORONTO

CHE1148 'Artificial Intelligence for Applied Chemistry and Chemical Engineering' • Jan 2025 – Pres.

CHE1147 'Chemical Data Science and Engineering' • Sep 2025 – Dec 2025

CHE222 'Process Dynamics: Modeling, Analysis and Simulation' • Jan 2025 – May 2025

- Conducted tutorials and practical laboratories on data science, machine learning, and numerical methods for chemical engineering.

### GLOBALINK MENTOR | MITACS

Mitacs Inc. • Toronto, Canada • Apr 2025 – Oct 2025

- Provided on-site mentorship to international undergraduate students during their 12-week stay in Canada with the Globalink Research Internship (GRI) program.

### RESEARCH INTERN | LABORATORY FOR METABOLIC SYSTEMS ENGINEERING

University of Toronto • Toronto, Canada • Jul 2023 – Oct 2023

- Conducted a computational research internship sponsored by the Mitacs GRI program.
- Developed a Python framework for the reconstruction of Genome-Scale Metabolic Models.

### ORGANIZING COMMITTEE | LATIN FOOD 2022

Mexican Association of Food Science (AMECA, A.C.) • Puebla, Mexico • Jun 2022 – Nov 2022

- Participated in the organization of one of Mexico's largest biotechnology congresses, leading the review of +500 abstracts of the presented research.

## PUBLICATIONS

R. A. Barghout, M. Kirby, A. Zheng, L. Chinas, M. Mohammadi, Z. Xu, B. Sanchez-Lengeling, and R. Mahadevan, "Machine Learning for Sequence-to-Function Approaches," in *Machine Learning and Big Data-enabled Biotechnology*, John Wiley & Sons, 2026, pp. 103-129.

## PRESENTATIONS

### CONFERENCE ON CONSTRAINT-BASED RECONSTRUCTION AND ANALYSIS

Potsdam, GER • 2026 • Poster presentation

San Diego, USA • 2024 • Oral & poster presentations

### AMERICAN INSTITUTE OF CHEMICAL ENGINEERS ANNUAL MEETING

San Diego, USA • 2024 • Oral presentation

### INTERNATIONAL CONFERENCE ON LEARNING REPRESENTATIONS (ICLR) 2025

Singapore • 2025 • Learning Meaningful Representations of Life Workshop

## AWARDS

### CONNAUGHT INTERNATIONAL SCHOLARSHIP FOR DOCTORAL STUDENTS

University of Toronto • 2024 – 2029

### SECIHTI SCHOLARSHIP FOR GRADUATE STUDIES

Mexican Secretary of Science, Humanities, and Innovation • 2024 – 2029